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ScienceDirect

Procedia Computer Science 165 (2019) 759–765

Procedia Computer Science

www.elsevier.com/locate/procedia

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING 2019, ICRTAC 2019

Object Detection In Underwater Acoustic Images Using Edge Based Segmentation Method

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Abstract

Image segmentation techniques play a vital role in partitioning the image into segments. These techniques are used for detecting the objects in the images. The real challenge in the acoustic image segmentation lies in differentiating the sea floor, sediments and the objects. Due to the typical characteristics and low resolution of the acoustic images, segmentation techniques such as region based, morphological based and edge based methods are not suitable. The sediments will also be traced along with the objects that makes the detection process a very difficult one. In this research paper, edge based segmentation method that uses the morphological operations for identifying the edges followed by a object tracing algorithm is proposed. The images used in this work are captured in real time using Edgetech 4125 Side scan sonar device. The acoustic images are first pre-processed using Wiener and Median filters and a morphological gradient is obtained by subtracting the morphologically dilated and eroded image. Next the edge map of the acoustic images are obtained using binarization method by which the object's boundary is made visible. Finally a Moore's object tracing algorithm is used for detecting the objects in the images.

Keywords: Acoustic; Edges; Object; Morphological; Segmentation; Sonar;

1. Introduction

Underwater exploration is a challenging task as it involves searching for things that are unusual and unexpected. It helps in managing the ocean resources. Image processing plays a major role in revealing the mystery and the wealth of deep sea. The acoustic imaging system helps to capture the images that are present under water such that it is useful for any further investigation and exploration.

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10.1016/j.procs.2020.01.015

Acoustic instruments are used for searching the surface of the ocean. Today, Sonar systems become a common acoustic-based technology for underwater exploration. SONAR (Sound Navigation And Ranging) is a technology used to detect and locate objects under the water. Side scan sonar [26] is one of such instruments used for detecting the objects in seafloor. The side scan sonar helps in accurate mapping of large sections of the seabed, locate pipeline or cable routes, obstructions and other features. Specifically, shipwreck location, mine hunting, downed aircraft search and lost cargo operations all require the use of side scan sonar.

The images of the seafloor are taken using sound as the source. This is due to the degradation in the optical visibility range in sea water which is caused by the presence of suspension, turbidity of water, unavailability of light source in deep sea. The sound waves are used to capture images in spite of turbid water and low visibility. The acoustic energy penetrates the mud and silt that causes turbidity and has the capability of capturing images in deep sea where human's presence is impossible. Hence the range of acoustic systems is greater than the optical systems. But the resolution capacity of acoustic imaging systems is lower than the optical image system.

Edge detection is a preprocessing technique that aims at identifying the boundary of the objects. There are many image processing techniques using various filters that detect the edges. The edges in the images are obtained by calculating the first and second order derivative [2][3]. These techniques may detect true edges if the foreground object is distinguishable from the background in an image. In case of texture images where the boundary is very different to differentiate, these filtering techniques may lead to false edges. The images used in this work are acoustic images obtained by side scan sonar. The acoustic images contain speckle noise caused by the instrument [9]. So there is a need for new edge detection technique that differentiates the object in the sea floor from the sediments and surface. This work aims at finding the true edges that are real objects using morphological techniques and then a boundary tracing algorithm is used for object detection.

2. Related Work

The acoustic images obtained from side scan sonar are texture images that contain seafloor, sediments and objects (both living and non living). Many researchers focus on detection and classification of the object lying under the sea bed. Many different works have been carried out for segmenting the acoustic images and to detect the objects. Some of the works similar to our proposed research work have been reviewed. In order to perform image segmentation, a combination of texture and spectral analysis [1] was proposed. For analysing the spectral features, Directional Filter Bank (DFB) was used and for deriving the texture features, Grey Level Co-occurrences Matrices (GLCM) was used. The acoustic images contains the texture features containing different regions. An active contour model is then applied to the original image. The segmentation method proposed by [4] readjusts the weights attached to each feature by modifying the feature selection step. The drawback of the method is the sensitivity to initialization.

A novel method was proposed by [6] for segmenting the side scan sonar images. This uses K-means clustering algorithm which is an unsupervised learning algorithm used to segment the image into various regions. The region based segmentation methods are well accepted ones for object detection. The authors [7] have done object detection in three steps. An object detection model is first used for positioning the objects and in next step, background location is determined using background prior. These steps are followed by region merging segmentation method. This method seems to be an automatic method for object detection in underwater images. Medical images are considered to be similar to underwater acoustic images as both are texture and gray scale images. Based on Tsallis and Shannon entropy a new edge detection algorithm was proposed for recognizing the human organs in the medical images. Due to the noise and geometric features [8] of the images, the edge detection techniques cannot be used for identifying the true edges.

Some methods perform image enhancement before detecting the objects from the images. As a preprocessing step in object detection dynamic brightness assignment is done [9]. The author [10] has done a comparative study on edge detection algorithms which were applied on noisy image using morphological filter. A work has been carried out by [11] for detecting the edges using the combination of mathematical morphological operators such as erosion, dilation, opening and closing.

Other than the texture images morphological operators have been generalized and used on complex signals [12]. Edge detection becomes more challenging when the objects remain in the dark regions. The traditional operators perform edge detection by first applying the gradient operators and then performing threshold. The method which uses pseudo top-hat transformation of the mathematical morphological edge detection algorithm [13] proves to detect the edges in the dark regions and also preserves distinguished features. In this method, thresholding is done using recursive quad tree decomposition scheme. Hidden Markov Chain (HMC) model along with non decimated wavelet have been used by [14] for detecting the edges in the images. EM (Expectation Maximization) algorithm has been used for training the model and Viterbi algorithm has been used for revealing the hidden state of each coefficient.

Many research works have been carried out for side scan sonar images including correction of brightness variation and patching gaps [15]. Due to the constantly changing attitude of the towfish the images are difficult to read. A voting based approach [16] applied directly to the edges recognizes the objects using priori known models. The drawback of this method is recognizing a single segment in the image. Detection, classification and identification of objects in side scan sonar [18][22] was done using six different screening algorithms. A marker based watershed transform [19] was used for detection of objects in salient regions and their cast shadows. Acoustic shadows provide useful information about the object's shape and are considered as salient regions. The work proposed by [20] has used mean shift clustering based segmentation technique for isolating the regions in an image. The reliability aware fusion of features were computed and used for different data set. For automatic detection of objects in underwater acoustic images obtained by forward looking sonar, the object's echo shape is segmented by removing the ground echo shape. This separated shape is compared to the simulated reference shape to know its orientation or to identify the shape using shape matching method.

A symmetric bank of Gabor filters are used for filtering the images and then a morphological closing operator [23] was applied on the filtered image. Then active contour model are used for segmenting areas with different textures. The advantage of this method is multiple texture regions are identified easily. This method doesn't use edge features for segmenting the regions. In order, to reduce the high computational effort and time, programmable logic technology was used by [24]. In this work, they have divided the image into three regions such as acoustic highlight, seafloor reverberation and acoustic shadow areas. They have applied adaptive methods for different regions. A 3D segmentation method [25] for underwater acoustic images has been done. They have used RANSAC (RANdom Sample And Consensus) algorithm through which a super quadric primitives are directly recovered from raw data without any pre-segmentation processing. Using evolutionary algorithms [29] have done 3D reconstruction of underwater scenes using DIDSON acoustic sonar image sequences.

In the literature, many researchers have used various segmentation algorithms for object detection. In the generic images, the objects were separated into background and foreground objects and were easily differentiated using region based and morphological based segmentation algorithms. But the object detection in the acoustic images are really challenging as they are texture images. It is very difficult to differentiate the objects from sea floor and sediments. In our research, we have used edge based segmentation method and object tracing methods for object detection.

3. Segmentation Techniques

The edges in the images can be detected using segmentation techniques. The segmentation techniques can be classified as region based and edge based segmentation.

3.1 Region based segmentation techniques

There are methods such as thresholding, region growing, region splitting and region merging that determines the edges in the objects. As the edges are the changes in the intensity of the pixels, the neighboring pixels are examined to determine the connected components. The region based methods considers the image as a whole and starts to evaluate all the pixels in the images.

3.2 Edge based segmentation techniques

The change in the intensity can be identified by differencing adjacent pixels. The process of finding the first order derivative can be done by convolving the filter to the image. Based on the values in the filters, the edge detectors are classified into Robert, Prewitt, Sobel and Canny. These edge detectors identify the edges in both horizontal and vertical directions. They use two different templates for determining the horizontal and vertical edges. The complete edge map is obtained by adding the horizontal and vertical edges.

Some of the edge detection operators such as Robert and Sobel don't perform edge linking. So the object boundary may not be visible. But the Canny edge detector identifies the edges and also performs the edge linking by smoothing the image, localizing the edges and thresholding for finding the true edges. The acoustic images that are texture images containing only seafloor and sediments. These are considered to be the connected components and it leads to clumsy edges.

4 Proposed Method for Edge Detection in Acoustic Images

The edge map is generally used for tracing the objects in an image. Identifying the true edges is the challenging task in detection. In acoustic images, this process is even more crucial since the sediments and the sea floor cannot be differentiated from the object in the obtained edges. The proposed method for edge detection first uses the Wiener filtering to remove the speckle noise. The speckle noise is the granular noise caused by the sonar which is used for acoustic image acquisition. The filtering process smoothes the image and preserves the high frequency components. Next step is smoothing the image using median filter which removes the small objects which is the sediments in our acoustic image.

Morphological processing is generally used for finding the shape of the objects by obtaining the local minimum and local maximum. This process includes erosion and dilation. The morphological gradient is used to obtain the edges in the images by using the difference of the dilated image and the eroded image. The resultant edge map is compared with the state of art edge detection techniques such as Canny and Sobel and used for tracing the objects in the images. The Figure 1 shows the block diagram for edge detection process.

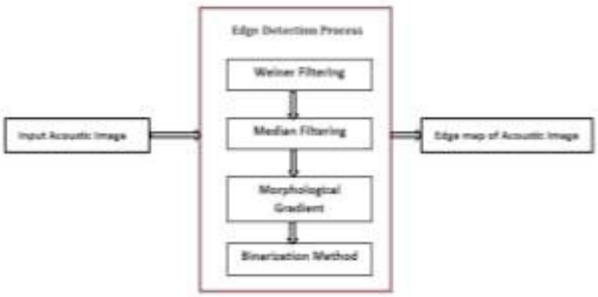

Figure 1: Block Diagram for Edge Detection Process

5. Results and Discussion

5.1 Experimental Results

The figures 2 shows the experimental results of the edge detection process for the input acoustic images 1 and 2. Figures 2.b) and c) represents the output of the preprocessing stage after applying wiener and median filters respectively. Figures 2.d) and e) indicates the output of Canny and Sobel edge detectors. The Figure 2.f) denotes the output of the proposed system. The output of both the Canny and Sobel operators reveals that they have detected false edges and ignored true edges. One of the characteristics of Canny edge detector is the edge linking property. It is obvious from the figure that the sediments and the sea floor also gets linked to create false edges. In Sobel, output true edges have been ignored. But the output of our proposed morphological based edge segmentation have detected only the true edges that is boundaries of the true objects and have avoided the edge linking.

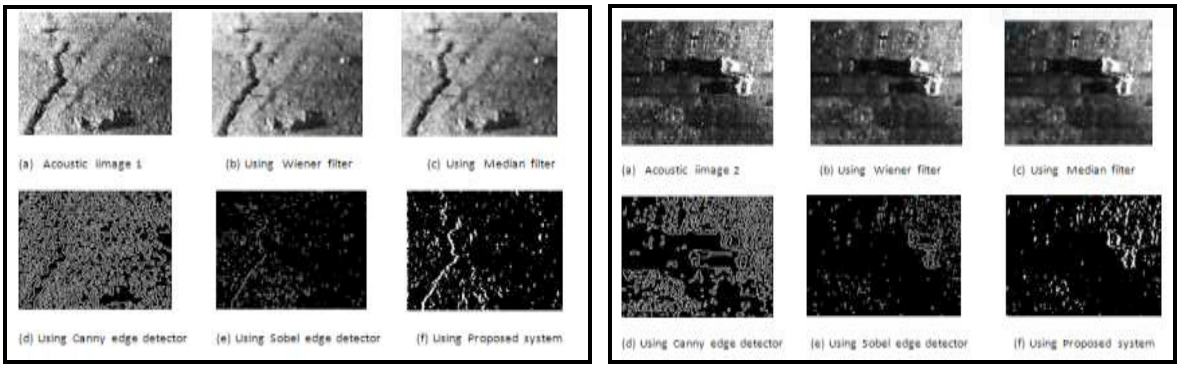

Figure 2: a) Acoustic Image1 b) Output of Wiener filter c) Output of Median Filter d) Output of Canny edge detector e) Output of Sobel edge operator f) Output of Proposed system

6 Boundary Tracing

Boundaries are the set of pixels in which the elements are edges in the binary image. The objects in the acoustic images are detected using the Moore's neighbor algorithm. Using this procedure, the boundary of the objects in the acoustic images are traced.

Procedure: Moore's neighbor algorithm

Input: Binary image with detected edges as white pixels
Output: Pixels forming the boundary of an object
1: Scan the pixels from leftmost corner of the image till a edge pixel is found
2: An edge pixel is considered as the starting point for tracing
3: Using eight neighborhood, the connected components for the starting pixel is determined.
4: A boundary is drawn over the connected pixels
5: The process is repeated till all the pixels are scanned

6.1 Experimental Results

Figure 3: Objects Detected in Acoustic images using edge based segmentation and boundary tracing method

7. Conclusion

This work aims at detecting the objects in the acoustic images. Before detecting the objects, the edges available in the images are identified. The speckle noise in the images is removed using Wiener filter and to still smooth the image, median filter was used. After preprocessing the image, edges are obtained by finding the morphological gradient. The morphological gradient is obtained by differencing the dilated image and the eroded image. The dilation process thickens the edges in the image whereas the erosion process shrinks the boundaries. When gradient is calculated only the true edges are retained and the false edges are ignored. The output of these steps leads to a binary image with edges. The objects in the acoustic images are detected using Moore's object tracing algorithm which uses the neighborhood operations to find the connected components. In visual perception the results show that the objects in the acoustic images are traced in a better way.

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